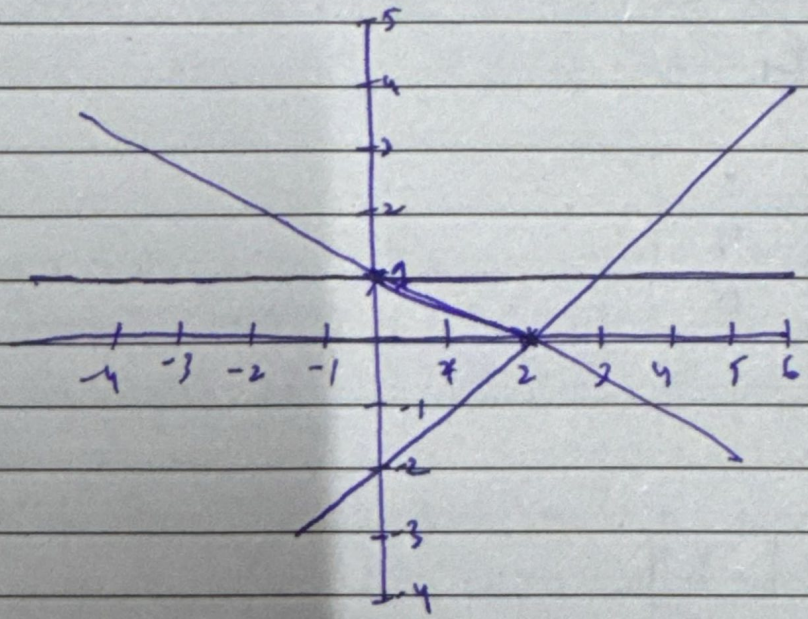
LA Assignment #01 Date:

Q1 $x+2y=2$, $y=-x/2+1$

$x-y=2$, $y=x-2$

$y=1$



There is no common point so the system is inconsistent (no solutions).

Date:

Q2

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 4 & 7 & 5 & 20 \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$R_3 \leftarrow R_3 + 2R_2$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 8 & 8 \end{array}\right]$$

$8z = 8$, $z = 1$

$y + 3(1) = 4$

$y = 1$

$2x + 3 + 1 = 8$

$2x + 4 = 8$

$2x = 4$

$x = 2$

$(x, y, z) = (2, 1, 1)$

pivot positions $(1,1), (2,2), (3,3)$

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Date:

Q3

part (i)

multiplications : $m \cdot n$ or (mn) each of the m rows require n multiplications to compute the dot product with n.

part (ii)

multiplications : $m \cdot n \cdot p$ or mnp matrix B has p column vectors of size n so the mn multiplications are repeated p times.

Q4 $A = \begin{bmatrix} 1 & 2 & 0 \\ 9 & 8 & 3 \\ 0 & b & 5 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 9R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 8-29 & 3 \\ 0 & b & 5 \end{bmatrix}$

$\det(A) = \det \begin{bmatrix} 8-29 & 3 \\ b & 5 \end{bmatrix} = 5(8-29) - 3b = 40 - 109 - 3b$

Row exchange : $9 = 4$, $b \neq 0$

Singular $109 + 3b = 40$

M.Z Zaver Product

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Q5

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 4 & 8 \\ -1 & 5 & 3 \end{bmatrix},$$

$$B = \begin{bmatrix} 4 & -2 & 1 & 0 \\ 2 & 4 & 1 & 3 \\ 5 & 6 & 8 & 2 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 2 & 8 \\ 3 & 4 & 2 \\ 5 & 6 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$(XY)^T = Y^T X^T$, so $(CB)^T = C^T B^T$

$$CB = \begin{bmatrix} 48 & 54 & 67 & 22 \\ 30 & 22 & 23 & 16 \\ 32 & 14 & 11 & 18 \\ 13 & 12 & 11 & 8 \end{bmatrix}, \quad B^T C^T = \begin{bmatrix} 48 & 30 & 32 & 13 \\ 54 & 22 & 14 & 12 \\ 67 & 23 & 11 & 11 \\ 22 & 16 & 18 & 8 \end{bmatrix} = (CB)^T$$

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(b)

$$B^TB = \begin{bmatrix} 45 & 30 & 46 & 16 \\ 30 & 56 & 50 & 24 \\ 46 & 50 & 66 & 19 \\ 16 & 24 & 19 & 13 \end{bmatrix}$$

$$(B^TB)(B^TB) = \begin{bmatrix} 5297 & 5714 & 6910 & 2522 \\ 5714 & 7112 & 7936 & 3086 \\ 6910 & 7936 & 9333 & 3437 \\ 2522 & 3086 & 3437 & 1362 \end{bmatrix}$$

(c) $BC = \begin{bmatrix} 3 & 6 & 28 \\ 22 & 32 & 27 \\ 65 & 86 & 54 \end{bmatrix}$, $BC+A = \begin{bmatrix} 6 & 6 & 29 \\ 24 & 36 & 35 \\ 64 & 91 & 57 \end{bmatrix}$

$$(BC+A)(BC+A) = \begin{bmatrix} 2036 & 2891 & 2037 \\ 3248 & 4625 & 3951 \\ 6216 & 8847 & 8290 \end{bmatrix}$$

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Q6

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1/2$

$R_2 \leftarrow R_2 + R_1$

$R_2 \leftarrow (2/3) R_2$

$$\left[\begin{array}{ccc|ccc} 1 & -1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & -2/3 & 1/3 & 2/3 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Next use $R_1 \leftarrow R_1 + 1/2 R_2$, $R_3 \leftarrow R_3 + R_2$, $R_3 \leftarrow 3/4 R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1/3 & 2/3 & 1/3 & 0 \\ 0 & 1 & -2/3 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & 1/2 & 3/4 \end{array}\right]$$

$R_1 \leftarrow R_1 + 1/3 R_3$ and $R_2 \leftarrow R_2 + 2/3 R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & 1/2 & 1/4 \\ 0 & 1 & 0 & 1/2 & 1 & 1/2 \\ 0 & 0 & 1 & 1/4 & 1/2 & 3/4 \end{array}\right]$$

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Q7 $AX = b$

$$\begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} -8/3 \\ 7/3 \\ 4/3 \end{bmatrix}$$

Q8 $A^{-1}(AB) = A^{-1}(AC) \Rightarrow (A^{-1}A)B = (A^{-1}A)C \Rightarrow IB = IC$

$B = C$ as required.

Q9

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = AC$$

$AB = AC$ but $B \neq C$. left cancellation fails for singular A.

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Date:

Q.00
$$\begin{cases} u+v+2w=2 \\ 2u+3v-w=5 \\ 3u+4v+w=c \end{cases}$$

$c=7$,

$(u,v,w) = (1-t, 1+t, t)$

Q.01

$x_2 - x_3 = 100$

$x_3 - x_4 = -500$

$x_1 - x_2 \leq 300$

$x_1 - x_4 = -100$

$$\left[\begin{array}{cccc|c} 0 & 1 & -1 & 0 & 100 \\ 0 & 0 & 1 & -1 & -500 \\ 1 & -1 & 0 & 0 & 300 \\ 1 & 0 & 0 & -1 & -100 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -100 \\ 0 & 1 & 0 & -1 & -400 \\ 0 & 0 & 1 & -1 & -500 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$(x_1, x_2, x_3, x_4) = (t+300, t, t-100, t+400)$, $t \geq 100$

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(b)
$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 0 & 50 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 & 450 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 750 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 600 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & -50 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

for all flows to be +ative we need
$0 \le s \le 750$ and $t \ge \max\{50, s-600\}$.
one valid flow occurs in $s=700$, $t=100$ giving:

$$(x_1, x_2, x_3, x_4, x_5, x_6, x_7) = (650, 350, 450, 400, 50, 300, 100)$$

Q2.

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 0 \end{bmatrix}$$

M.Z Zawar Product

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Q3

$$A = \begin{bmatrix} 3 & 5 & -1 \\ 4 & -1 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

direct substitution at $(-1, 2, 4)$

$$W_1 = 3(4) + 5(2) - 4 = 3$$
$$W_2 = 4(-1) + -2 + 4 = -2$$
$$W_3 = 3(-1) + 2(2) - 4 = -3$$

$$A \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 3 \end{bmatrix}$$

Q4

$$D_2 \begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{bmatrix}$$

$$(k+1)\begin{vmatrix} k-3 & 4 \\ k+1 & k \end{vmatrix} - (k-1)\begin{vmatrix} 2 & 4 \\ 5 & k \end{vmatrix} + 7\begin{vmatrix} 2 & k-3 \\ 1 & k+1 \end{vmatrix}$$

$$= (k+1)(k^2 - 4k - 4) - (k-1)(2k - 20) + 7(17 - 3k)$$
$$= k^3 - 8k^2 - 10k + 95$$

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Q5. $D_2 = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix}$

$R_2 \leftarrow R_2 - 5R_1$, $R_3 \leftarrow R_3 + R_1$,
$R_4 \leftarrow R_4 - 2R_1$, $R_4 \leftarrow R_4 - 12R_2$
$R_4 \leftarrow R_4 + 36R_3$

$$\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & 13 \end{bmatrix}$$

$D_2 = 1 \cdot 1 \cdot (-3) \cdot (-13) = 39$

Q6 $\det(A) = \det(A^T)$

$$\det(A) = 4\begin{vmatrix} 2 & -3 \\ 1 & 5 \end{vmatrix} - (-2)\begin{vmatrix} 0 & -3 \\ -1 & 5 \end{vmatrix} - \begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix}$$

$$= 4(13) - 2(-3) - 2 = 56$$

$$A^T = \begin{bmatrix} 4 & 0 & -1 \\ 2 & 2 & 1 \\ -1 & -3 & 5 \end{bmatrix}$$

$$\det(A) = 4\begin{vmatrix} 2 & 1 \\ -3 & 5 \end{vmatrix} - \begin{vmatrix} 2 & 2 \\ -1 & -3 \end{vmatrix}$$

$$= 4(13) - (-4) = 56$$

M.Z Zawar Product

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